MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER
NAND GATES, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-0038510/20C(USAF), dated 1 May 1974.

Page 3

TABLE I, Device type 01, test 27 thru 33: Add "*" to each test (all 7 tests).

Page 7

TABLE I, Device type 03, tests 5 thru 12, pin 6: Delete "2.4 V" and substitute "GND" (all 8 tests).

TABLE I, Device type 03, test 19, pin 12: Delete "5.5" and substitute "5.5 V".

TABLE I, Device type 03, tests 25 and 26, pin 8: Delete "open" and substitute "5.5 V".

Page 9

TABLE I, Device type 04, test 13: Delete "±.10" and substitute "±10".

Page 10

TABLE I, Device type 04, test 37, pins 6, 7, 9, and 10: Delete "5.5 V" and substitute "open" (4 places).

TABLE I, Device type 04, test 38: Delete "GND" under pins 1 and 2 and substitute "open". Delete "5.5 V" under pins 9 and 10 and substitute "open".

TABLE I, Device type 04, tests 39 and 40, pins 1 and 2: Delete "GND" and substitute "open".

TABLE I, Device type 04, minimum limit column, tests 38 thru 40: Add -3 mA as minimum limit (all 3 tests).

Pages 14 and 15

TABLE II, Switching characteristics, Device types 01, 02, 03, 04, and 05: Delete and substitute new tables as printed on pages 2 and 3 of this amendment.

Page 21

FIGURE 4, note 1: Delete "and $T_{THL} = T_{TLH} \leq 60ns$".

Page 22

FIGURE 5, note 3: Delete "T_A = 25 ° C" and substitute "T_A = 25°C".

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-42)

FSC 5962

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TABLE II. (Device type 01) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits Min	Limits Max	Units
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	99	ns
				125°C, -55°C	3	154	
Propagation delay time low to high level output	t_{PLH}			25°C	3	66	ns
				125°C, -55°C	3	99	

TABLE II. (Device type 02) Switching characteristics.

Characteristic	Symbol	Conditions 1/	Test figure	Temperature	Limits Min	Limits Max	Units
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	66	ns
				125°C, -55°C	3	99	
Propagation delay time low to high level output	t_{PLH}			25°C	3	66	ns
				125°C, -55°C	3	99	

1/ Test each gate.

TABLE II. (Device type 03) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits Min	Limits Max	Units
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	66	ns
				125°C, -55°C	3	99	
Propagation delay time low to high level output	t_{PLH}			25°C	3	66	ns
				125°C, -55°C	3	99	

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TABLE II. (Device type 04) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5.0 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	66	ns
				125°C, -55°C	3	99	
Propagation delay time low to high level output	t_{PLH}			25°C	3	66	ns
				125°C, -55°C	3	99	

TABLE II. (Device type 05) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	C_L = 50 pF R_L = 4 kΩ ±5% V_{CC} = 5.0 V	4	25°C	3	66	ns
				125°C, -55°C	3	99	
Propagation delay time low to high level output	t_{PLH}			25°C	3	66	ns
				125°C, -55°C	3	99	

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